measurements, you confirm that its mass is, indeed, 15 trillion times that of the Sun, you see that it is spinning very slowly, and you compute from these data that the circumference of its horizon is 29 light-years. Here, at last, is a hole whose vicinity you can explore while experiencing bearably small tidal forces and rocket accelerations! The safety of the exploration is so assured that you decide to take the entire starship down instead of just a capsule.

Before beginning the descent, however, you order your crew to photograph the giant quasar overhead, the trillions of stars that orbit Gargantua, and the billions of galaxies sprinkled over the sky. They also photograph Gargantua's black disk below you; it is about the size of the sun as seen from Earth. At first sight it appears to blot out the light from all the stars and galaxies behind the hole. But looking more closely, your crew discover that the hole's gravitational field has acted like a lens[30] to deflect some of the starlight and galaxy light around the edge of the horizon and focus it into a thin, bright ring at the edge of the black disk. There, in that ring, you see several images of each obscured star: one image produced by light rays that were deflected around the left limb of the hole, another by rays deflected around the right limb, a third by rays that were pulled into one complete orbit around the hole and then released in your direction, a fourth by rays that orbited the hole twice, and so on. The result is a highly complex ring structure, which your crew photograph in great detail for future study.

The photographic session complete, you order Kares to initiate the starship's descent. But you must be patient. The hole is so huge that, accelerating and then decelerating at 1 *g*, it will require 13 years of starship time to reach your goal of 1.0001 horizon circumferences!

As the ship descends, your crew make a photographic record of the changes in the appearance of the sky around the starship. Most remarkable is the change in the hole's black disk below the ship: Gradually it grows larger. You expect it to stop growing when it has covered the entire sky below you like a giant black floor, leaving the sky overhead as clear as on Earth. But no; the black disk keeps right on growing, swinging up around the sides of your starship to cover everything except a bright, circular opening overhead, an opening through which you see the external Universe (Figure P.4). It is as though you had entered a cave and were plunging deeper and deeper, watching the cave's bright mouth grow smaller and smaller in the distance.